Ratios average different things

Three equally probable dataset cases.
One criterion; three units; M = 3.

Case 1: P = 1/3

g = (1, 2, 2)

(A, N) = (2, 2)

A/N = 1

N/M = 2/3

Case 2: P = 1/3

g = (1, 2, 3)

(A, N) = (0, 3)

A/N = 0

N/M = 1

Case 3: P = 1/3

g = (1, 1, 1)

(A, N) = (0, 0)
A/N: NA
No disclosure
N/M = 0

Pool counts

$(2 + 0 + 0)/(2 + 3 + 0) = 2/5$

Rate weights: (2/5, 3/5, 0).

Silent case contributes zero counts.

Average nonsilent rates

$(1 + 0)/2 = 1/2$

Conditional weights: (1/2, 1/2).

Exclude silence; keep its rate NA.

Impute zero, then average

$(1 + 0 + 0)/3 = 1/3$

Case weights: (1/3, 1/3, 1/3).

Extra convention: NA → 0.

P(N > 0) = 2/3; mean N/M = 5/9.
A/N is undefined when N = 0.
Different estimands and weights.